

Chomsky Normal Form

CSE 211 (Theory of Computation)

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Adapted from slides by
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Chomsky Normal Form

- Every nonempty CFL without ϵ has a grammar G in which all productions are in one of two simple forms, either:
 - ❶ $A \rightarrow BC$, where A , B , and C , are each variables, or
 - ❷ $A \rightarrow a$, where A is a variable and a is a terminal.
- Further, G has no useless symbols.
- Such a grammar is said to be in **Chomsky Normal Form**, or **CNF**.

Noam Chomsky

- “the father of modern linguistics” - wiki
- Linguist, philosopher, cognitive scientist, historian, social critic, and political activist
- Developed the theory of transformational grammar
- Author of many books and articles
 - Anti-war essay “The Responsibility of Intellectuals”
 - Criticism of media in “Manufacturing Consent”



Testing Membership in a CFL

- There is an efficient technique based on the idea of “dynamic programming”
- The algorithm is known as the **CYK Algorithm**.
 - Cocke-Younger-Kasami algorithm
- It starts with a CNF grammar $G = (V, \Sigma, R, S)$ for a language L .
- The input to the algorithm is a string $w = a_1 a_2 \dots a_n$ in Σ^* . In $O(n^3)$ time, the algorithm constructs a table that tells whether w is in L .

Testing Membership in a CFL

X_{15}					
X_{14}	X_{25}				
X_{13}	X_{24}	X_{35}			
X_{12}	X_{23}	X_{34}	X_{45}		
X_{11}	X_{22}	X_{33}	X_{44}	X_{55}	
a_1	a_2	a_3	a_4	a_5	

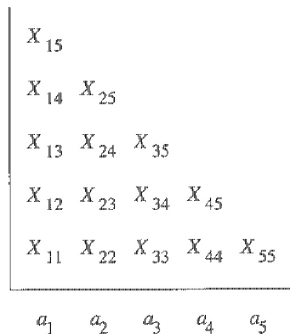
Figure 7.12: The table constructed by the CYK algorithm

Testing Membership in a CFL

X_{15}				
X_{14}	X_{25}			
X_{13}	X_{24}	X_{35}		
X_{12}	X_{23}	X_{34}	X_{45}	
X_{11}	X_{22}	X_{33}	X_{44}	X_{55}
a_1	a_2	a_3	a_4	a_5

- We construct a triangular table.

Testing Membership in a CFL



- The horizontal axis corresponds to the positions of the string $w = a_1 a_2 \dots a_n$.

Testing Membership in a CFL

X_{15}				
X_{14}	X_{25}			
X_{13}	X_{24}	X_{35}		
X_{12}	X_{23}	X_{34}	X_{45}	
X_{11}	X_{22}	X_{33}	X_{44}	X_{55}
a_1	a_2	a_3	a_4	a_5

- The table entry X_{ij} is the set of variables A such that $A \xRightarrow{*} a_i a_{i+1} \dots a_j$.

Testing Membership in a CFL

X_{15}				
X_{14}	X_{25}			
X_{13}	X_{24}	X_{35}		
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a_1	a_2	a_3	a_4	a_5

- We are interested in whether S is in the set X_{1n} , because that is the same as saying $S \xRightarrow{*} w$, i.e., w is in L .

Testing Membership in a CFL

- X_{ij} is the set of variables A such that $A \rightarrow a_i$ is a production of G .
- In order for A to be in X_{ij} , we must find variables B and C , and integer k such that:
 - i $i \leq k < j$.
 - ii B is in X_{ik} .
 - iii C is in $X_{k+1,j}$.
 - iv $A \rightarrow BC$ is a production of G .

Example

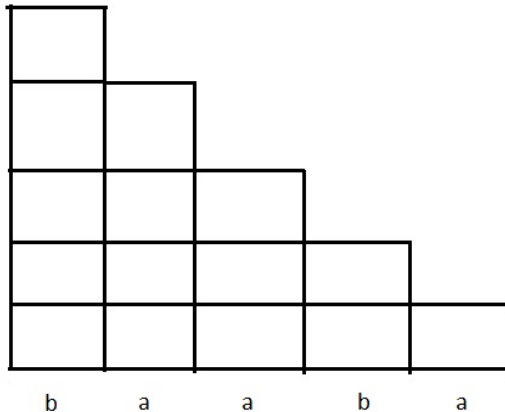
The following are the productions of a CNF grammar G

- $S \rightarrow AB|BC$
- $A \rightarrow BA|a$
- $B \rightarrow CC|b$
- $C \rightarrow AB|a$

We shall test for membership in $L(G)$ the string *baaba*.

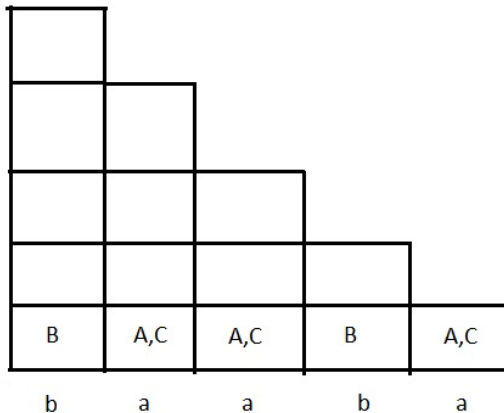
Example

- $S \rightarrow AB|BC$
- $A \rightarrow BA|a$
- $B \rightarrow CC|b$
- $C \rightarrow AB|a$



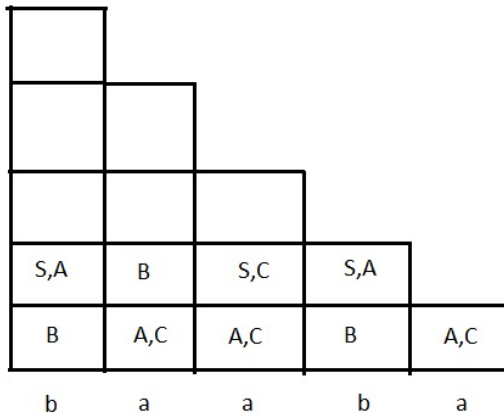
Example

- $S \rightarrow AB|BC$
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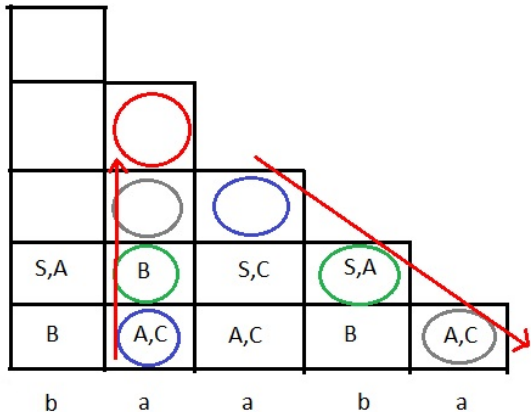
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- $S \rightarrow AB|BC$
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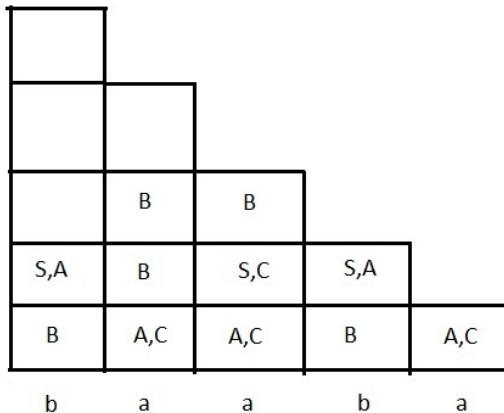
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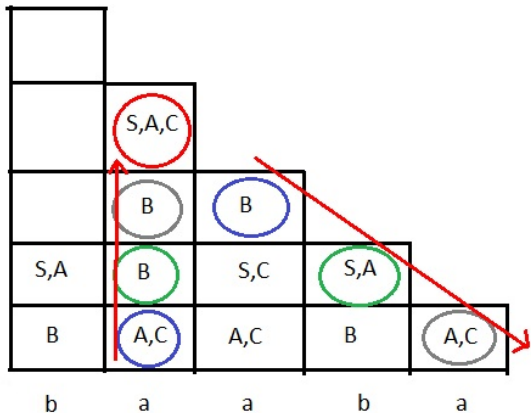
Example

- $S \rightarrow AB|BC$
- $A \rightarrow BA|a$
- $B \rightarrow CC|b$
- $C \rightarrow AB|a$



Example

- $S \rightarrow AB|BC$
- $A \rightarrow BA|a$
- $B \rightarrow CC|b$
- $C \rightarrow AB|a$



Example

$\{S, A, C\}$				
-	$\{S, A, C\}$			
-	$\{B\}$	$\{B\}$		
$\{S, A\}$	$\{B\}$	$\{S, C\}$	$\{S, A\}$	
$\{B\}$	$\{A, C\}$	$\{A, C\}$	$\{B\}$	$\{A, C\}$
b	a	a	b	a

Figure 7.14: The table for string *baaba* constructed by the CYK algorithm

Chomsky Normal Form

- To convert a grammar to its Chomsky Normal Form, we need to make a number of preliminary simplifications:
 - i We must eliminate *useless symbols*, those variables or terminals that do not appear in any derivation of a terminal string from the start symbol.
 - ii We must eliminate ϵ -*productions*, those of the form $A \rightarrow \epsilon$ for some variable A .
 - iii We must eliminate unit productions, those of the form $A \rightarrow B$ for variables A and B .

Chomsky Normal Form

Theorem

Any context-free language is generated by a context-free grammar in Chomsky normal form.

Converting to CNF

Perform the following steps in this order:

- i Eliminate ϵ productions
- ii Eliminate unit productions
- iii Eliminate useless symbols
- iv Convert to CNF
 - a Arrange that all bodies of length 2 or more consist only of variables
 - b Break bodies of length 3 or more into a cascade of productions, each with a body consisting of two variables

Eliminating Useless Symbols

- Two things a symbol has to be able to do to be useful
 - We say X is generating if $X \xRightarrow{*} w$ for some terminal string w . Note that every terminal is generating since w can be that terminal itself
 - We say X is reachable if there is a derivation $S \xRightarrow{*} \alpha X \beta$ for some α and β
- If a symbol is not useful, it is *useless*

Example

- Consider the grammar

$$\begin{aligned} S &\rightarrow AB \mid a \\ A &\rightarrow b \end{aligned}$$

- Find generating and reachable symbols using induction
- B is not generating

$$\begin{aligned} S &\rightarrow a \\ A &\rightarrow b \end{aligned}$$

- A is not reachable

$$S \rightarrow a.$$

Eliminating ϵ -productions

- Discover variables that are *nullable*
 - A variable A is nullable if $A \xRightarrow{*} \epsilon$
- If A is nullable, then whenever A appears in a production body, say $B \rightarrow CAD$, A might or might not derive ϵ . We make two versions of the production
 - one without A in the body $B \rightarrow CD$ which corresponds to the case where A would have been used to derive ϵ
 - and the other with A still present $B \rightarrow CAD$
- If language contains ϵ , add $S \rightarrow \epsilon$ where S is the start symbol

Example

- Consider the grammar

$$S \rightarrow AB$$

$$A \rightarrow aAA \mid \epsilon$$

$$B \rightarrow bBB \mid \epsilon$$

- A, B and S are nullable
- Production 1 becomes

$$S \rightarrow AB \mid A \mid B$$

- Production 2 becomes

$$A \rightarrow aAA \mid aA \mid aA \mid a$$

Example

- Similarly

$$B \rightarrow bBB \mid bB \mid b$$

- So, the grammar after eliminating ϵ -productions is

$$S \rightarrow AB \mid A \mid B$$

$$A \rightarrow aAA \mid aA \mid a$$

$$B \rightarrow bBB \mid bB \mid b$$

- Since S is nullable add $S \rightarrow \epsilon$

Eliminating unit productions

- A unit production is a production of the form $A \rightarrow B$ where both A and B are variables
- Identify *unit pairs*
 - A pair (A, B) is called unit pair if $A \xRightarrow{*} B$ using only unit productions
- For each unit pair (A, B) , add all the productions $A \rightarrow \alpha$, where $B \rightarrow \alpha$ is a nonunit production. Note that $A = B$ is possible in that way. Only the non-unit productions remain

Example

- Consider the grammar

$$\begin{aligned}
 I &\rightarrow a \mid b \mid Ia \mid Ib \mid I0 \mid I1 \\
 F &\rightarrow I \mid (E) \\
 T &\rightarrow F \mid T * F \\
 E &\rightarrow T \mid E + T
 \end{aligned}$$

- Find the unit pairs. $(E, E), (T, T), (F, F), (I, I)$ are unit pairs by zero steps

- (E, E) and the production $E \rightarrow T$ gives us unit pair (E, T) .
- (E, T) and the production $T \rightarrow F$ gives us unit pair (E, F) .
- (E, F) and the production $F \rightarrow I$ gives us unit pair (E, I) .
- (T, T) and the production $T \rightarrow F$ gives us unit pair (T, F) .
- (T, F) and the production $F \rightarrow I$ gives us unit pair (T, I) .
- (F, F) and the production $F \rightarrow I$ gives us unit pair (F, I) .

Example

- The productions to be added/kept

Pair	Productions
(E, E)	$E \rightarrow E + T$
(E, T)	$E \rightarrow T * F$
(E, F)	$E \rightarrow (E)$
(E, I)	$E \rightarrow a \mid b \mid Ia \mid Ib \mid I0 \mid I1$
(T, T)	$T \rightarrow T * F$
(T, F)	$T \rightarrow (E)$
(T, I)	$T \rightarrow a \mid b \mid Ia \mid Ib \mid I0 \mid I1$
(F, F)	$F \rightarrow (E)$
(F, I)	$F \rightarrow a \mid b \mid Ia \mid Ib \mid I0 \mid I1$
(I, I)	$I \rightarrow a \mid b \mid Ia \mid Ib \mid I0 \mid I1$

Example

- The resulting grammar

$$E \rightarrow E + T \mid T * F \mid (E) \mid a \mid b \mid Ia \mid Ib \mid I0 \mid I1$$

$$T \rightarrow T * F \mid (E) \mid a \mid b \mid Ia \mid Ib \mid I0 \mid I1$$

$$F \rightarrow (E) \mid a \mid b \mid Ia \mid Ib \mid I0 \mid I1$$

$$I \rightarrow a \mid b \mid Ia \mid Ib \mid I0 \mid I1$$

Converting to CNF

- The grammar has had its ϵ -productions, unit productions and useless symbols removed
- Our tasks are to
 - Arrange that all bodies of length 2 or more consist only of variables
 - Break bodies of length 3 or more into a cascade of productions, each with a body consisting of two variables

Example

- Consider the grammar

$$E \rightarrow E + T \mid T * F \mid (E) \mid a \mid b \mid Ia \mid Ib \mid I0 \mid I1$$

$$T \rightarrow T * F \mid (E) \mid a \mid b \mid Ia \mid Ib \mid I0 \mid I1$$

$$F \rightarrow (E) \mid a \mid b \mid Ia \mid Ib \mid I0 \mid I1$$

$$I \rightarrow a \mid b \mid Ia \mid Ib \mid I0 \mid I1$$

Example

- Eight terminals $a, b, 0, 1, +, *, (, \text{ and })$, appears in a body that is not a single terminal
- We must introduce eight new variables, corresponding to these terminals, and eight productions in which the new variable is replaced by its terminal

$$\begin{array}{llll} A \rightarrow a & B \rightarrow b & Z \rightarrow 0 & O \rightarrow 1 \\ P \rightarrow + & M \rightarrow * & L \rightarrow (& R \rightarrow) \end{array}$$

Example

- We introduce these productions, and replace every terminal in a body that is other than a single terminal by the corresponding variable

$$\begin{aligned}
 E &\rightarrow EPT \mid TMF \mid LER \mid a \mid b \mid IA \mid IB \mid IZ \mid IO \\
 T &\rightarrow TMF \mid LER \mid a \mid b \mid IA \mid IB \mid IZ \mid IO \\
 F &\rightarrow LER \mid a \mid b \mid IA \mid IB \mid IZ \mid IO \\
 I &\rightarrow a \mid b \mid IA \mid IB \mid IZ \mid IO \\
 A &\rightarrow a \\
 B &\rightarrow b \\
 Z &\rightarrow 0 \\
 O &\rightarrow 1 \\
 P &\rightarrow + \\
 M &\rightarrow * \\
 L &\rightarrow (\\
 R &\rightarrow)
 \end{aligned}$$

Example

- Introduce variables to break bodies of length 3 or more

$$E \rightarrow EC_1 \mid TC_2 \mid LC_3 \mid a \mid b \mid IA \mid IB \mid IZ \mid IO$$

$$T \rightarrow TC_2 \mid LC_3 \mid a \mid b \mid IA \mid IB \mid IZ \mid IO$$

$$F \rightarrow LC_3 \mid a \mid b \mid IA \mid IB \mid IZ \mid IO$$

$$I \rightarrow a \mid b \mid IA \mid IB \mid IZ \mid IO$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$Z \rightarrow 0$$

$$O \rightarrow 1$$

$$P \rightarrow +$$

$$M \rightarrow *$$

$$L \rightarrow ($$

$$R \rightarrow)$$

$$C_1 \rightarrow PT$$

$$C_2 \rightarrow MF$$

$$C_3 \rightarrow ER$$